



SUJET STATAPP 302 - 2026-2027



Sujet proposé par : Guillot Gilles - g.guillot@woah.org
Organisme : World Organisation for Animal Health (WOAH)
Catégorie : Microéconomie Appliquée
Sujet :
Animal Health as Market and Trade Intelligence

Description du sujet :

The Animal Health Econometrics group at WOAH's Data Integration and Analytics Department is currently drafting the programme of work ?Animal Health as Market and Trade Intelligence.?

Animal diseases affect meat supply on commodity markets, which can in turn affect food security.

The OECD-FAO Agricultural Outlook 2025-2034 states that:

?Animal disease outbreaks continue to disrupt the meat sector significantly [?] The continuous evolution and spread of animal diseases such as HPAI virus, ongoing ASF, resurgence of Foot and Mouth Disease (FMD), and the New World Screwworm (NWS) cast uncertainties for the medium term.?

This student project supports the design of that work programme , building proofs of concept for a few of its components.

Données du sujet :

The project will draw on public meat-market data (volume and price) at three levels of aggregation, alongside WOAH's own disease-event data.

Global / cross-country

- FAO FAOSTAT ? production, trade and food balance data by country, back to 1961.
- FAO Meat Price Index / GIEWS Food Price Monitoring ? monthly global meat price index; country- and some market-level price monitoring.
- OECD-FAO Agricultural Outlook ? national and regional projections and historical series for production, consumption, trade and prices, by meat type.
- USDA FAS PSD Online / Livestock and Poultry: World Markets and Trade ? country-level production, supply and trade.
- World Bank Commodity Markets ("Pink Sheet") and IMF Primary Commodity Prices ? global meat/livestock price indices.
- UN Comtrade ? bilateral trade volumes by HS code, aggregatable to regional/global.

National

- USDA ERS Livestock and Meat Domestic Data, Meat Price Spreads, and International Trade Data (US).
- USDA AMS Market News and USDA NASS QuickStats (US, down to state/county ? the most granular US source).
- Eurostat Livestock and Meat, and the EC Agri-food Data Portal weekly meat market prices (EU member states).
- Statistics Canada, China NBS/MARA price monitoring, Australia MLA, UK AHDB ? national series, several with provincial/state breakdowns.

Sub-national

- USDA AMS Market News (specific terminal markets, plants, regions).
- USDA NASS QuickStats (state, county).
- China NBS/MARA (provincial), Statistics Canada (provincial), Brazil CEPEA/Esalq (state-level), Australia MLA (saleyard-level).

Disease/outbreak data

- WOAHS WAHIS (World Animal Health Information System) for outbreak timing, location and disease events, to be matched against the price/volume series above.

Modèles envisagés du sujet :

- Event-study / structural-break models: abnormal price or futures-return reaction around an outbreak announcement date (cf. Paiva-style event studies; Chow/Bai-Perron break tests).
- Threshold or regime-switching models (TAR/STAR): nonlinear price response above a disease-severity or media-attention threshold (cf. the ASF/AI South Korea threshold study).
- GARCH / multivariate GARCH-DCC: time-varying volatility and cross-market volatility transmission (cf. the international beef-price volatility study).
- VAR / Granger causality and impulse-response analysis: lead-lag relationships between producer, wholesale and retail prices, and across countries.
- Panel fixed-effects or difference-in-differences: price/trade impact of trade bans or SPS measures, exploiting variation in ban timing and geographic scope across countries or US states/counties (cf. Padilla, Baker & MacLachlan, 2025).
- Gravity models: effect of SPS measures on bilateral trade flows (cf. Crivelli & Gröschl, 2016).
- ****Assessment of model predictive accuracy**** (out-of-sample validation, cross-validation across outbreak episodes).
- ****Communication of results**** to OECD-TAD colleagues.

Bibliographie du sujet :

Market outlook

- ****[G]**** OECD/FAO (2025), **OECD-FAO Agricultural Outlook 2025-2034**, Meat chapter. - Source of the rationale quote above; medium-term projections for global meat production, consumption, trade and prices, flagging HPAI, ASF, FMD and New World Screwworm as ongoing sources of uncertainty for the meat sector.

Theoretical foundation

- ****[T]**** McInerney, Howe & Schepers (1992), *"A framework for the economic analysis of disease in farm livestock,"* Preventive Veterinary Medicine. - The foundational "loss-expenditure frontier" model distinguishing production losses from control expenditure; underpins nearly all later disease-economics work.

Empirical price-impact studies by disease type

****African Swine Fever (ASF)****

- ****[E]**** Acosta et al. (2023), *"The ripple effect of animal disease outbreaks on food systems: ASF on the Chinese pork market,"* Food Policy. - Retail pork prices rose ~2x farmgate price rises during 2018-19 China ASF (asymmetric transmission); spillover raised beef/chicken retail prices too.
- ****[E]**** *"Threshold Effects of Infectious Disease Outbreaks on Livestock Prices: ASF and Avian Influenza in South Korea"* (2021). - Nonlinear/threshold-autoregressive response of pork prices to ASF-related news/social-media mentions; AI showed only a linear (no threshold) effect on chicken prices.
- ****[G]**** USDA ERS, *"How China's African Swine Fever Outbreaks Affected Global Pork Markets"* (ERR-326, 2023). -

China lost ~27.9 million metric tons of pork output (2018-2021); domestic prices roughly doubled; record imports offset only ~1/5 of the loss.

****Avian Influenza (HPAI/AI)****

- ****[E]**** Mitchell, Thompson & Malone (2024), ****"Biological lags and market dynamics... HPAI impacts on U.S. egg prices,"** Food Policy. - HPAI raised U.S. weekly retail egg prices ~5.3% on average (2022-23); models ignoring biological/production lags misstate direction and magnitude of consumer-surplus effects.

****Foot-and-Mouth Disease (FMD)****

- ****[E]**** ****"Market Impact of Foot-and-Mouth Disease Control Strategies: A UK Case Study"** (PMC5585142). - Links an epidemiological model (EXODIS) to a partial-equilibrium market model (FAPRI-UK); finds the largest driver of value-of-output loss is the export-market closure/price drop, not culling itself.

- ****[E]**** ****"Cost-Benefit Analysis of an FMD Disease-Free Zone in Punjab, Pakistan"** (PMC8416473). - Median benefit-cost ratio of only ~1.03 (range spans below 1); loss of herd (market) value accounts for 75-87% of total outbreak losses at farm level, dwarfing direct control costs.

****PPR (Peste des Petits Ruminants)****

- ****[E]**** ****"The Economic Impact of Eradicating Peste des Petits Ruminants: A Benefit-Cost Analysis"** (PMC4764769). - 15-year global eradication program: cost US\$2.26bn vs. benefits US\$76.5bn (BCR 33.8); annual PPR losses estimated at US\$1.475bn.

****BVD (Bovine Viral Diarrhoea)****

- ****[E]**** ****"Cost-benefit analysis and potential trade effects of the bovine viral diarrhoea eradication programme in Styria, Austria"** (ScienceDirect/Preventive Veterinary Medicine). - Voluntary phase had positive BCR (1.18); compulsory phase alone was cost-negative (BCR 0.16) but coincided with significant cattle export volume (+42-47%) and producer price increases (+5-14%) - market/trade gains can offset a negative direct cost-benefit ratio.

Trade bans, export restrictions & sanitary (SPS) barriers

- ****[E]**** Crivelli & Gröschl (2016), ****"The Impact of Sanitary and Phytosanitary Measures on Market Entry and Trade Flows,"** The World Economy. - Gravity-model evidence: SPS measures are an obstacle to market entry, but conditional on entry, some SPS types (product-characteristic standards) raise trade flows while others (conformity/testing measures) hamper entry.

- ****[G]**** USDA ERS Amber Waves, ****"Embargoes From Past Avian Flu Outbreaks Reduced Demand for U.S. Chicken Meat"** (2022). - 2014-15 embargoes cut U.S. poultry exports ~145,000 tons and cost producers \$332-472m; after the 2004 outbreak exports fell >20% yet domestic prices rose slightly - an example of divergent price responses across outbreaks.

- ****[E]**** Padilla, Baker & MacLachlan (2025), ****"The impact of HPAI trade restrictions on U.S. poultry exports in 2022-23,"** Applied Economic Perspectives and Policy. - State-level restrictions cut real export value ~60% vs. more narrowly targeted county-level restrictions (~16% decline) - geographic granularity of bans matters greatly for price/trade impact.

- ****[G]**** FAO (livestock/TAD economics review, 2016). - FMD causes the greatest trade restriction among transboundary diseases; ~70% of world trade in FMD-susceptible animals/meat (2009) came from officially FMD-free zones. Distinguishes demand-shock price collapses (e.g., 2004 Cambodia H5N1 chicken/egg price crash) from

supply-shock price spikes, and notes the 1997-98 Rift Valley Fever export ban cost the Somali region ~25% of GDP. Cites Paarlberg et al. (2002): a combined epidemiological-economic US model estimated a potential US\$14bn farm-income loss from an FMD outbreak.

Price volatility & value-chain transmission

- **[E]** International beef price volatility transmission study (2020), GARCH/DCC models, Jan 2006-May 2020. - International beef prices Granger-cause regional prices (Georgia, Tajikistan, US); volatility linkages strengthen sharply in crisis periods, including disease-driven disruptions; recommends trade-restriction "shielding" of domestic markets during emergencies.
- **[E]** "Media reports amplify market risk in pork prices..." Frontiers in Sustainable Food Systems (2025). - Media coverage of animal epidemics raises pork price volatility (effect strongest in consumption regions and via commercial vs. official outlets); consumer online attention partially mediates the effect; government policy interventions dampen it.

Methodology, institutional frameworks & control-cost trade-offs

- **[E]** "Livestock health and disease economics: a scoping review" (Frontiers in Veterinary Science, 2023). - Reviews 192 papers; identifies a literature gap on the global economic burden of disease and how it's distributed across value chains, plus a lack of full (indirect, national-level) FMD impact studies.
- **[G]** FAO Animal Production and Health Guidelines No. 18, "Economic Analysis of Animal Diseases." - Standardized methodological framework for assessing market/price and value-chain cost distribution of disease.
- **[G]** WOA, **GBADs** (Global Burden of Animal Diseases) programme - gbads.woah.org. - Standardized global data/methodology for measuring visible costs (mortality, productivity, veterinary spend) and hidden losses (environmental, gender, livelihood) from animal disease; production animals = ~40% of global agricultural value, and ASF alone caused a loss equal to 10% of global pork supply.

Notation du sujet :

Commentaires :